

TECHNOLOGY AT YOUR SERVICE

Powering slums with rejected laptop batteries

IBM India, in association with RadioStudio, a hardware R&D firm, has come up with a technology that can power slums in India and other developing countries using discarded laptop batteries.

At a conference in San Jose, USA, researchers presented their invention and stated that 70 per cent of discarded lithium-ion laptop batteries have so much energy left that these can power an LED light on a regular basis for at least four hours each day throughout an year.

Vikas Chandan, lead research scientist at Smarter Energy Group, said that the most costly component required for this process is the battery, and if this innovation is successful, then it can come from the trash. The rechargeable device has been dubbed UrJar, which has been made using parts of discarded lithium-ion laptop batteries to power an LED light and a mobile charger. UrJar means a combination of energy and box.

Storage units were extracted from multiple battery samples and re-combined to form battery packs. Then, charging ports were added and a rigorous testing process followed. The device was offered to five users in Bengaluru slums. On-the-field testing was successful as participants found it safer, cheaper and easier-to-use. Users suggested adding more features like FM radio and some wires, which would not be vulnerable to rats. The cost of the device is expected to be around ₹ 600.

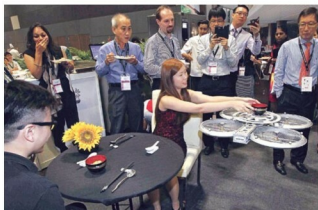
Researchers have suggested more than one variant of UrJar, which can be charged by solar energy, and batteries other than lithium-ion ones can also be used. IBM has proposed that this technology should be offered to developing countries for free.

Flying robots to replace waiters in restaurants

We have seen and heard of robots doing household chores, from being a baby-sitter to a driver, but a robotic waiter is a new concept. Infinitum-Serve, the world's first commercial robot, aims to save human resource required for serving food. It is designed and developed by a small company, Infinitum Robotics.

The robot will be launched in Timbre Group's five restaurants in Singapore this year. The current set-up cost for five robotic systems is a low seven-figure. However, the cost can be reduced as demand and production increases.

Both companies, Infinitum Robotics and Timbre Group, are trying to raise government funds to reduce deployment costs. This technology can help save the expenditure on manpower. The cumbersome task of tak-



Flying robot serving food

ing orders, managing chefs, serving food and drinks can be taken up by Infinitum-Serve.

This spoon does not spill food

Google is working on developing a spoon that does not spill food. This spoon will help people suffering from tremors or Parkinson's disease. The spoon has passed clinical-testing and has shown positive results—it has reduced shaking by 76 per cent.

The company has developed algorithms that prevent the spoon from shaking. The technology senses when a hand is shaking and balances the spoon automatically to prevent spilling. Google's Life Sciences division is behind this project. The device even has a fork attachment.

Google got into this niche market with the acquisition of the start-up Lift Labs, a company responsible for the product. The Lifeware spoon technology will help more than ten million people worldwide.

Google's co-founder, Sergey Brin, whose mother is suffering from essential tremors, has donated over US\$ 50 million for the research. There are devices, like rock-er knives, weighted utensils, pen grips, that have been developed in the past to help people with tremors, but the technology has never been used in this way before.

TV frequencies to deliver Wi-Fi

German scientists at Karlsruhe Institute of Technology (KIT) have proposed that some TV frequencies are available for free, and these can be extended as wireless area networks (WLAN) or Wi-Fi. These will help reduce dependency on frequencies for mobile communication.

Low-range TV frequencies can penetrate obstacles like walls. Researchers plan to extend these frequencies for free communication to include lower ranges and to increase transmission power. As transmission power will automatically be adapted to prevent any kind of